Rich Mixture

The following lists some of the possible combinations of exhaust gas values and the most likely causes.

CO	CO2	HC	O2	A/F Ratio too Rich: possible problems, conditions or causes
High	Low	Low to Moderate	Low	• Rich fuel mixture, • Leaking injectors, • Incorrect carburetor adjustment, • Power valve leaking, • Choke operating rich (closed), • Float level too high, Air filter dirty, • EVAP canister purge system faulty, • PCV system problem, • ECM malfunctioning, • Crankcase contaminated with raw fuel
Moderate to High	Low	Low to Moderate	Low	• All of the above but with catalytic converter operating correctly.
High	Low	High	High	• Rich mixture with ignition misfire
Low	High	Low	Low	**Good combustion efficiency and catalytic converter working properly!**

Note: High CO, Low CO2 , VERYHigh HC(>1000) and VERYHigh O2(>5) is sometimes indicator of WRONG ENGINE VALVE TIMING. Simply explained, what enter's the engine (Air+Fuel), get's out only partially burned because of the poor mechanical condition of the engine. More about engine problems...here.

Causes of Excessive Exhaust Emissions:

As a general rule, excessive HC, CO, and NOx levels are most often caused by the following conditions:

- Excessive HC results from ignition misfire or misfire due to excessively lean or rich air/fuel mixtures
- Excessive CO results from rich air/fuel mixtures
- Excessive NOx results from excessive combustion temperatures

When troubleshooting these types of emissions failures, you will be focusing on identifying the cause of the conditions described above. For example, to troubleshoot the cause of excessive CO emissions, you need to check all possible causes of too much fuel or too little air (rich air fuel/ratio). The following lists of causes will help familiarize you with the sub-systems most often related to excessive CO, HC and NOx production.

Causes of Excessive HC - Hydrocarbons

As mentioned, high hydrocarbons is most commonly caused by engine misfires. The following list of problems could cause high HC levels on fuel injected vehicles. As with any quick reference, there are other less likely causes that may not be included in the list. Here are some of the more common causes:

- Ignition system failures
    1. faulty ignition secondary component
    2. faulty individual primary circuit on distributorless ignition system
    3. weak coil output due to coil or primary circuit problem
- Excessively lean air/fuel mixture
    1. leaky intake manifold gasket
    2. worn throttle shaft
- Excessive EGR dilution
    1. EGR valve stuck open or excessive EGR flow rate
    2. EGR modulator bleed plugged
- Restricted or plugged fuel injector(s)
- Closed loop control system incorrectly shifted lean
- False input signal to ECM
    1. incorrect indication of load, coolant temp., O2 content, or throttle position
- Exhaust leakage past exhaust valve(s)
    1. tight valve clearances
    2. burned valve or seat
- Incorrect spark timing
    1. incorrect initial timing
    2. false input signal to ECM
    3. worn piston rings or cylinder walls
- Insufficient cylinder compression
- Carbon deposits on intake valves

Causes of Excessive CO - Carbon Monoxide

High carbon monoxide levels are caused by anything that can make the air/mixture richer than "ideal". The following examples are typical causes of rich mixtures on fuel injected vehicles:

- Excessive fuel pressure at the injector(s)
- Leaking fuel injector(s)
- Ruptured fuel pressure regulator diaphragm
- Loaded/malfunctioning EVAP system (two speed idle test)
- Crankcase fuel contamination (two speed idle test)
- Plugged PCV valve or hose (two speed idle test)
- Closed loop control system incorrectly shifted rich
- Excessive combustion blow-by
- False input signal to ECM - incorrect indication of load, coolant temp., O2 content, or throttle position

Note: It should be pointed out that due to the reduction ability of the catalytic converter, increases in CO emissions tend to reduce NOx emissions. It is not uncommon to repair a CO emissions failure and, as a result of another sub-system deficiency, have NOx increase sufficiently to fail a loaded-mode transient test.

Causes of Excessive NOx - Oxides of Nitrogen

Excessive oxides of nitrogen can be caused by anything that makes combustion temperatures rise. Typical causes of high combustion temperature on fuel injected vehicles include:

- Cooling system problems
    1. insufficient radiator airflow
    2. low coolant level
    3. poor cooling fan operation
    4. thermostat stuck closed or restricted
    5. internal radiator restriction
- Excessively lean air/fuel mixture
    1. leaky intake manifold gasket
    2. worn throttle shaft
- Closed loop control system incorrectly shifted lean
- Improper oxygen sensor operation

1. slow rich to lean switch time
2. rich biased 02 sensor voltage

- Improper or inefficient operation of EGR system
  1. restricted EGR passage
  2. EGR valve inoperative
  3. EGR modulator inoperative
  4. plugged E or R port in throttle body
  5. faulty EGR VSV operation
  6. leaky/misrouted EGR hoses
- Improper spark advance system operation
  1. incorrect base timing
  2. false signal input to ECM
  3. improper operation of knock retard system
- Carbon deposits on intake valves

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